chemical changes 1

Name:.. Date:......................................

Which of the following is correctly balanced?

A. $Fe + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + H_2$
B. $2 Fe + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + H_2$
C. $2 Fe + 3 H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3H_2$
D. $Fe + 3 H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3H_2$

Correct Answer: A B C D

Explanation:

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ROYAL SOCIETY OF CHEMISTRY

Which symbol equation for the reaction between sodium carbonate (Na_2CO_3) and hydrochloric acid (HCl) correctly represents the word equation below?
Sodium carbonate (solid) + Hydrochloric acid (solution) → Sodium chloride (solution) + Carbon dioxide (gas) + Water (liquid)

A $Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + 2CO_2(g) + H_2O(l)$

B $Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$

C $Na_2CO_3(s) + HCl(aq) \rightarrow NaCl(aq) + CO_2(g) + H_2O(l)$

D $2Na_2CO_3(s) + 4HCl(aq) \rightarrow 4NaCl(aq) + CO_2(g) + 2H_2O(l)$

Correct Answer: A B C D

Explanation:

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4 Which compound, when mixed with aqueous barium nitrate, does **not** form a white precipitate?

A ammonium carbonate
B dilute sulfuric acid
C silver nitrate
D sodium carbonate

Correct Answer: A B C D

Explanation:

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3 Which ionic equation describes a redox reaction?

A $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

B $2H^+(aq) + CO_3^{2-}(aq) \rightarrow CO_2(g) + H_2O(l)$

C $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

D $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Correct Answer: A B C D

Explanation:

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2 An aqueous solution of a compound **X** reacts with

- aqueous zinc chloride to form a white precipitate which dissolves when **X** is in excess,
- aluminium sulfate solution to form a white precipitate which is insoluble when **X** is in excess.

What is the identity of **X**?

A ammonia

B barium chloride

C silver nitrate

D sodium hydroxide

Correct Answer: A B C D

Explanation:

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C5 Ionic Equations & Precipitation Reactions

Sodium hydroxide is used to test for metal ions.

$FeCl_{2(aq)} + NaOH_{(aq)} \rightarrow Fe(OH)_{2\,(s)} + 2NaCl_{(aq)}$

Identify the spectator ion(s).

A Na^+

B Fe^{2+} and Cl^-

C Na^+ and Fe^{2+}

D Na^+ and Cl^-

Correct Answer: A B C D

Explanation:

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C5 Ionic Equations & Precipitation Reactions

A solution of sodium iodide reacts with a solution of silver nitrate.

Solid sodium iodide does not react with **solid** silver nitrate.

Identify the result of the solutions reacting and explain why the solutions react, but the ions do not.

	Result	Explanation
A	Yellow precipitate	Ions cannot move in a solid.
B	Cream precipitate	Ions cannot move in a solid.
C	Yellow precipitate	Ions are in fixed positions in a solid.
D	Cream precipitate	Ions are in fixed positions in a solid.

Correct Answer: A B C D

Explanation:

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Strontium is in group 2 of the periodic table. What is the formula of the salt formed when nitric acid is neutralised by strontium hydroxide?

A $SrNO_3$

B $Sr(NO_3)_2$

C $SrNO_2$

D $Sr(NO_2)_2$

Correct Answer: A B C D

Explanation:

The equations show what happens when four compounds dissolve in water. Which one of the named compounds is not an alkali?

A Ammonia: $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

B Barium hydroxide: $Ba(OH)_2(s)$ + water $\rightarrow Ba^{2+}(aq) + 2OH^-(aq)$

C Hydrogen sulfide: $H_2S(aq) \rightleftharpoons H^+(aq) + HS^-(aq)$

D Methylamine: $CH_3NH_2(aq) + H_2O(l) \rightleftharpoons CH_3NH_3^+(aq) + OH^-(aq)$

Correct Answer: A B C D

Explanation:

A student tested two solutions and recorded these results:

- pH = 1 for dilute hydrochloric acid.
- pH = 3 for a sample of vinegar.

How much lower was the concentration of hydrogen ions in vinegar compared to that of dilute hydrochloric acid?

A 2 times lower

B 3 times lower

C 100 times lower

D 1000 times lower

Correct Answer: A B C D

Explanation:

C5 Strong & Weak Acids

Dilute sulfuric acid of pH 4 is added to a test-tube containing water.

What could the pH of the resulting solution?

A pH = 8

B pH = 6

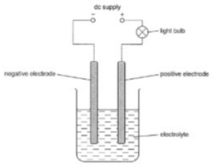
C pH = 4

D pH = 2

Correct Answer: A B C D

Explanation:

C5 Strong & Weak Acids



Two electrolysis experiments are set up, one using a weak acid and one using a strong acid of the same concentration.

The bulb glows much brighter with the strong acid.

Explain why in terms of ionisation and H^+ ions.

	Ionisation	**H^+ ions**
A	Strong acid fully ionises	Strong acid has higher concentration
B	Weak acid fully ionises	Strong acid has higher concentration
C	Strong acid fully ionises	Weak acid has higher concentration
D	Weak acid fully ionises	Weak acid has higher concentration

Correct Answer: A B C D

Explanation:

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C5 Strong & Weak Acids

Sodium hydroxide (NaOH) is a **strong base**.

What does this mean?

A NaOH fully dissociates when it is dissolved in water.

B NaOH cannot be neutralised.

C An aqueous solutions of NaOH contain equal concentrations of H^+ and OH^- ions.

D NaOH has a low pH.

Correct Answer: A B C D

Explanation:

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C5 Strong & Weak Acids

Identify the **strong** acid.

A $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$

B $HCOOH \rightleftharpoons H^+ + HCOO^-$

C $CuCO_3 \rightarrow CuO + CO_2$

D $HF \rightarrow H^+ + F^-$

Correct Answer: A B C D

Explanation:

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Explain why magnesium chloride must be molten or dissolved in water to be electrolysed.

A. Because ions can move to the electrodes and carry charge

B. Because atoms can move the electrodes and carry charge

C. Because electrodes carry charge and need to move for electrolysis

D. Because electrons can move to the electrodes and carry charge

Correct Answer: A B C D

Explanation:

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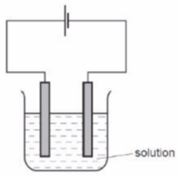
9 The diagram shows the electrolysis of a concentrated aqueous solution containing both copper(II) ions and sodium ions.



Which metal is deposited at the negative electrode and why?

	metal deposited	reason
A	copper	copper is less reactive than sodium
B	copper	copper is more reactive than hydrogen
C	sodium	copper is less reactive than hydrogen
D	sodium	copper is more reactive than sodium

Correct Answer: A B C D

Explanation:

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7 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

Which equation represents the reaction taking place at the anode (positive electrode) in this electrolysis?

A $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

B $SO_4^{2-}(aq) \rightarrow SO_2(g) + O_2(g) + 2e^-$

C $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

D $4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$

Correct Answer: A B C D

Explanation:

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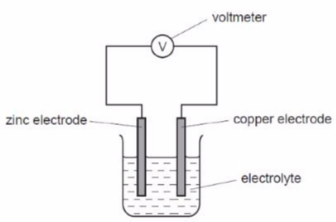
3 A simple cell is shown below.



Which statement about the process occurring when the cell is in operation is correct?

A Cu^{2+} ions are formed in solution.

B Electrons travel through the solution.

C The reaction $Zn \rightarrow Zn^{2+} + 2e^-$ occurs.

D Zinc increases in mass.

Correct Answer: A B C D

Explanation:

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An electrolysis apparatus is set up using two copper electrodes and aqueous copper(II) sulfate as the electrolyte. The electrolysis is carried out for 30 minutes. Predict what happens to the mass of each electrode during this time.

A — Mass of the anode – stays the same; mass of the cathode - decreases

B — Mass of the anode – increases; mass of the cathode - decreases

C — Mass of the anode – decreases; mass of the cathode - increases

D — Mass of the anode – increases; mass of the cathode – stays the same

Correct Answer: A B C D

Explanation:

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Electrolysis of an aqueous solution produced bubbles of gas at both electrodes. The gases were collected and tested using a lighted splint and damp blue litmus paper. The results are shown below:

Electrode	Test with lighted splint	Test with damp blue litmus paper
Anode	The flame went out	Went red and then white
Cathode	Made a pop noise	No change

Which of these compounds could have been the substance dissolved in the solution?

A lead nitrate

B potassium chloride

C silver chloride

D sodium hydroxide

Correct Answer: A B C D

Explanation:

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